

Smoothing Adoption Model

Analysis Results Briefing

Forecast-ready invoice timing signals built from 682 historic projects
Using an Empirical-Bayes shrunk weighted scorecard model

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01 | Analysis Definition - Invoice Smoothing

A project is treated as smoothed when its Last Date Invoiced is later than its First Date Invoiced — i.e. invoicing spread > 0 days.

Not Smoothed



First & last invoice on the same day (spread = 0).
Revenue recognised immediately.

Smoothed



Invoice dates span multiple days or months (spread > 0).
Revenue phased across the invoicing window.

Impact



Affects cashflow forecasting, period revenue recognition, and pipeline accuracy if not modelled.

Key question: Which future projects are likely to be smoothed, and by how many days?

02 | Data & Methodology

1

Collect historic projects

682 projects with first/last invoice dates across all categories, types, products, and accounts.

2

Define smoothing

Smoothed = spread days > 0. Unsmoothed = first and last invoice on same day.

3

Apply mature cohort filter

Projects whose first invoice is < 180 days old are excluded (right-censored). Mature cohort: 568 projects.

4

Calculate group rates

Smoothing probability and average spread computed for each category, type, product, and account.

5

Apply shrinkage ($k = 20$)

Group rates are pulled toward the portfolio average to prevent small groups from dominating forecasts.

6

Weight signals for scoring

Category 30% · Product 30% · Account 30% · Type 10% — combined into a single smoothing probability.

03 | Portfolio-Level Results

682

Total Projects

Full historic dataset

568

Mature Cohort

≥180 days since first invoice

36.8%

Smoothed Rate

Mature cohort probability

78 d

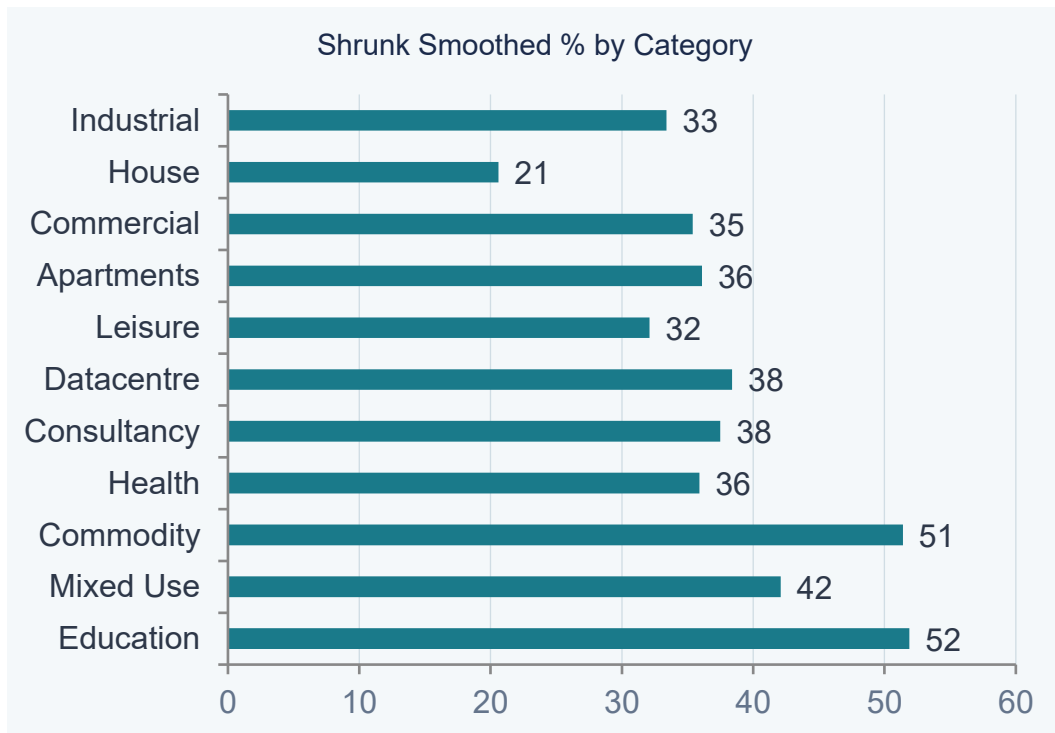
Expected Spread

Portfolio average (all projects)

Smoothed-project invoice spread distribution (mature cohort)

Metric	All Projects	Mature Cohort	365-day Sensitivity
Average spread (all)	68 days	78 days	99 days
Average spread (smoothed)	191 days	213 days	257 days
Median spread (smoothed)	91 days	112 days	162 days
P75 spread (smoothed)	225 days	256 days	367 days
P90 spread (smoothed)	522 days	567 days	631 days

04 | Category Signals



Top categories to watch

Education

52% | 104 d

Adopt smoothing by default

Consultancy

37% | 102 d

Model smoothing scenario

Apartments

36% | 84 d

Model smoothing scenario

Commercial

35% | 62 d

Flag / light smoothing

House

21% | 39 d

No default smoothing

Education Refurbishment is the standout sub-segment: 59% shrunk rate, 117-day expected spread — strong default smoothing signal.

05 | Product & Account Signals

Top Products

Product	Smoothed %	Exp. Days
EPS/PIR Torch On (Prebonded)	59.9%	162
Torch On PIR (Prebonded)	64.4%	176
ROCKWOOL HardRock Multi-Fix	42.9%	121
Tissue Faced PIR (Prebonded)	39.7%	136
Torch On PIR	48.9%	97
Foil Faced PIR	46.2%	108
ROCKDeck	48.2%	114
VIPs	43.0%	88

Key Accounts

Account	Smoothed %	Exp. Days
Garland	70.3%	160
Rockwool Limited	47.4%	145
Moy Materials (UK) Ltd	47.3%	142
Axter Limited	49.3%	120
ICB (Waterproofing) Ltd	30.7%	105
Rockwool Ltd (Bauder)	33.0%	106
Acclaim Contracts Limited	49.4%	151
Langley Waterproofing	43.6%	78

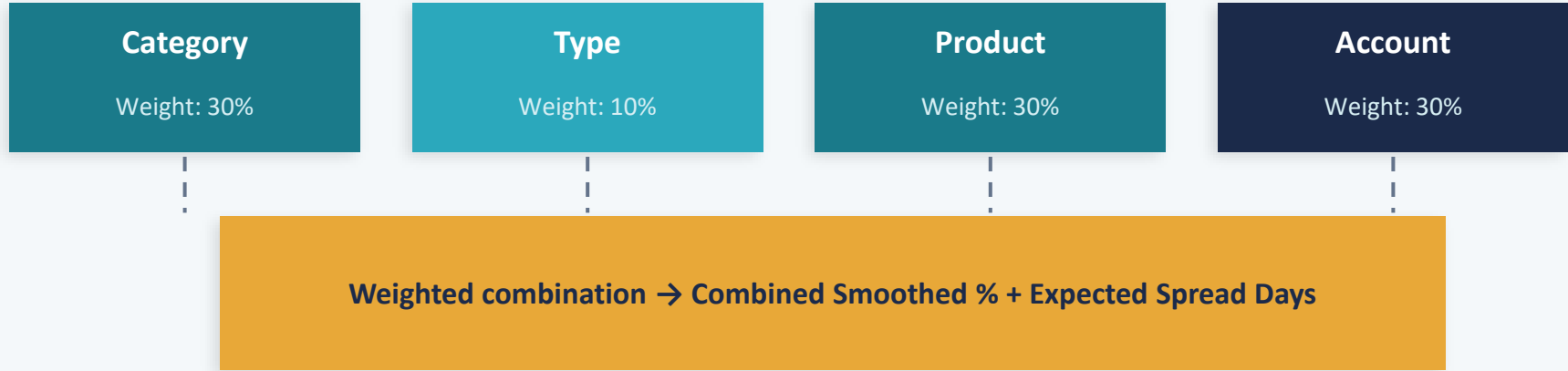
Garland stands out with a 70% smoothed rate and 160-day expected spread — the strongest single-account signal in the dataset.

06 | High-Priority Adoption List

Groups the model recommends treating with default smoothing today (mature projects ≥ 11 , confidence Medium–High)

Dimension	Group	Smoothed %	Exp. Spread	Confidence	Treatment
Product	Torch On PIR (Prebonded)	64.4%	176 d	Medium	Adopt smoothing by default
Product	EPS/PIR Torch On (Prebonded)	59.9%	162 d	Low	Adopt smoothing by default
Account	Garland	70.3%	160 d	Medium	Adopt smoothing by default
Account	Rockwool Limited	47.4%	145 d	Medium	Adopt smoothing by default
Account	Moy Materials (UK) Limited	47.3%	142 d	Medium	Adopt smoothing by default
Product	Tissue Faced PIR (Prebonded)	39.7%	136 d	Medium	Adopt smoothing by default
Product	ROCKWOOL HardRock Multi-Fix (DD)	42.9%	121 d	High	Adopt smoothing by default
Account	Axter Limited	49.3%	120 d	High	Adopt smoothing by default
Category+Type	Education Refurbishment	59.1%	117 d	High	Adopt smoothing by default

07 | Future Project Scoring



Unknown values fall back to the global mature portfolio average (36.8% | 78 days)

Output: Risk Band & Treatment

≥ 55%	45–55%	35–45%	< 35%
Very High	High	Moderate	Low
Adopt smoothing by default	Adopt smoothing by default	Model smoothing scenario	Flag / no default smoothing

08–09 | Caveats, Summary & Recommendations

Key Caveats

- ▶ No contract value data — model predicts timing tendency, not revenue-weighted smoothing impact.
- ▶ Right-censoring addressed by mature cohort, but 365-day sensitivity shows rates could be ~38% if extended further.
- ▶ Small groups (Low / Very Low confidence) have high shrinkage — signals are weak and require more data.
- ▶ Type alone is a weak differentiator (New Build vs Refurbishment both ~36–38%); use as secondary signal only.

Recommended Actions

Now

Apply default smoothing to the 9 high-priority groups on Slide 8 when building forecasts.

Now

Use the Future Scoring sheet to score all incoming projects at pipeline entry.

Soon

Add contract / invoice value to enable revenue-weighted smoothing impact, not just count-based.

Later

Incorporate PM, forecast period, and expected project duration for a richer multi-factor model.

Recommended enhancement: Add contract value, planned invoice profile and expected project duration for a revenue-weighted model.